

Prolog Programming for Artificial Intelligence

4th Edition

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July 14, 2025

1 Introduction to Prolog

1.1 Defining relations by facts

1.1

- (a) `?- parent( jim, X).`
`false.`
- (b) `?- parent( X, jim).`
`X = pat.`
- (c) `?- parent( pam, X), parent( X, pat).`
`X = bob.`
- (d) `?- parent( pam, X), parent( X, Y), parent( Y, jim).`
`X = bob,`
`Y = pat.`

1.2

- (a) `?- parent( X, pat).`
`X = bob.`
- (b) `?- parent( liz, X).`
`false.`

```
(c) ?- parent( X, pat), parent( Y, X).  
    X = bob,  
    Y = pam ;  
    X = bob,  
    Y = tom.
```

1.2 Defining relations by rules

1.3

```
(a) happy( X) :- hasachild( X).
```

```
(b) hastwochildren( X) :-  
    hasachild( X),  
    parent( X, Y),  
    sister( Y, Z).
```